**Welcome to the Boettcher Lab!**

# *Group Philosophy and Guidelines* ***2025***

This manual was inspired by manuals from the Sanford Lab, the Genna Lab, the McNeil Lab, the Aly Lab, and the Cook Lab.

# Table of Contents

I. 3
X. 3
A. 3
B. 4
C. 7
II. Joining the Boettcher Laboratory
III. 10
A. 10
B. 10
C. Google Drives / 10
D. 10
E. 10
IV. 10
A. 10
B. 11
C. 11
V. Time off and 11
A. 11
B. 11

# I. Philosophy for an Effective Research Team

## X. Preamble

As the leader of our research enterprise, there are three major drivers that influence how I try to manage our group:

1. Desire to educate scientists and engineers who will be highly competent once they leave, which means learning how to identify and tackle difficult unstructured problems effectively, efficiently and creatively, while productively communicating the results.

2. The practical need to use resources effectively to do excellent science important for our society – and that will be judged externally as high impact – so our team can continue our research with sufficient funding to make real progress.

3. The desire to help lab members be healthy and happy while part of our team, and also have successful careers in the long term.

-Shannon Boettcher

---

Below is an outline of a philosophy that was developed by the entire Boettcher group since 2020, as well as with input from a large number of alumni, to help us to balance these three, sometimes conflicting, drivers and other priorities of the students.

## A. Fostering individual and inter-personal balance (set healthy expectations)

1. **Schedule and prioritize self-care and work-life harmony.** Grad school is difficult. Burn-outs are real and the wider culture of self-care and work-life balance is lacking. You are no good to anyone else if you aren't good to yourself first. One can often be more productive with a well-scheduled efficient work day than a disorganized long workday. Cultivate good time-management practices. Don't compare yourself to others. Everyone has different life demands and works different hours. Develop outside interests, put those in your calendar, and stick to them. Often you will have to put in long hours, work over the weekend, or schedule instrument time in the evenings, but don't overdo it. Don't feel guilty for taking a vacation and disconnecting. Respect and support lab members who are caregivers and parents of young children. Take advantage of counseling services available on campus. Talk with your peers about self-care, mental health, and time management strategies.

2. **Strive for honest communication.** Be honest with your advisor and make it clear what you and/or your team want and why. This includes communicating your successes and failures. Research is highly non-linear and communication throughout can expedite iterating on experiments and research ideas. Do not avoid your PI. Try to understand other perspectives and realities. Completing a PhD program is demanding and formative. Being a PI is also extremely demanding. PIs are under constant pressure to procure funding, publish extensively, and manage university administrative issues and politics – all with no-one to take over their duties when needed. This reality, however, is not an excuse to not hold a PI to a high standard for mentorship, nor is it an excuse for PIs not to recognize the real challenges students face in their studies.

3. **Practice empathy and actively address issues before they become serious.** Fellow students are often the best suited and most aware of the issues their peers face. Create a culture where those who are proactive in caring and supporting those who are struggling are celebrated, thanked, etc. Almost everyone has large "ups and downs" in graduate school. Sometimes the downs can be really low and

everyone needs a strong peer group to get through this. If you see someone struggling check in with them and offer to help! Be intentional to check in on all group members, especially if some are working in lab spaces with less day-to-day interactions with others.

4. **Learn how to say "no", even to your PI.** You do not have to do everything, all the time. Avoid burnout by practicing saying "no" and coming up with creative solutions that *reduce busy work and increase efficiency*. Think about whether a request is better suited to someone else's expertise, career goals, schedule etc. Your PI's first choice for a job may not be the best choice necessarily. Show genuine gratitude when others help you. Learn to set healthy expectations both of yourself and with others.

5. **Create a culture of inclusivity, empowerment, and change.** Group leadership means making the changes that need to be made to improve the team function. Every student must feel supported in driving positive change in the group culture/dynamic as well as practical aspects. Regularly bring your suggestions to improve the group function to group meeting. View change as an experiment. Be willing to experiment in the social and organization dynamic in the group to find the best structure, which will evolve with the personnel and their needs. Don't be complacent. Recognize that every person has 20+ years of personal history that they bring to the table. They may also have a different cultural background than you and their previous and current experiences may be dramatically different from yours. Actively provide a seat at the table for and promote members of groups from all backgrounds. Practice active listening and empathy.

6. **Try to participate in activities organized by the lab** (outings, hikes, happy hours). Such activities help you to get to know your colleagues better and form strong bonds. Also try to attend presentations and practice talks given by your colleagues to support them and give your feedback. It's important to show interest and care towards your lab mates. Encourage others to do the same.

## B. Research Culture

1. **Collaborate**. "Credit is not a conserved quantity" – Steve Chu (Nobel Prize Winner in Physics). Shannon prefers including as co-authors all who contributed time, expertise, intellectual input, and/or collected data on manuscripts. This incentivizes a culture of collaboration and generates self-reinforcing positive energy. For example, include newer students in projects near publication and in the paper-writing process. As new students develop in their 2nd and 3rd year and start to publish their own work, include former mentors who contributed perhaps only intellectually as co-authors. However, there are many philosophies regarding authorship. Critically, **communicate with your collaborators** at the outset and throughout the research process regarding their contributions and appropriation of credit. As first author, think critically about who contributed to every aspect of your project and decide for each paper in consult with your PI the authorship and credit assignment.

2. **Avoid territorialism.** There are, literally, an infinite number of interesting and important research questions. If you feel there is effort duplication that is unproductive in the group, bring up the issue in sub-group or to your PI so that the responsibilities can be clearly delineated and new areas developed as needed. You can't and shouldn't plan your entire PhD. Welcome new additions to your group. They are new resources to help you solve problems. Integrate them into your team and work to publish with them collaboratively as fast as possible while doing good science. If the team's goal is to drive the science forward as effectively as possible, then everyone wins in the end. If someone else has a good idea that you want to try, just ask them first and offer to collaborate. Simultaneously, respect the boundaries between projects and specific requests from your coworkers about working on individual ideas. Be deliberate and specific in conversations about project boundaries.

3. **Work to have the group's interests in mind.** Consider offering to do routine things for others if you are already doing it for yourself, to save time. For example, making electrode substrates is a time-consuming activity, but you can make a lot in one shot. Ask your lab mates if they need those electrodes as well, make extra, and leave them in the shared drawer. Or if you make aqua regia solution to clean your Pt electrode, offer to clean other Pt electrodes that your lab mates use. Share papers relevant to others. It will minimize waste and save everyone's time. Everybody will appreciate you contributing to shared lab efforts.

4. **Embrace challenge.** Learn how to distinguish between being reasonably conservative with your time and energy, and being complacent or "lazy". Research will be hard, but lean into it. Doing hard things will make you a better scientist. Remember, from afar, things look more difficult than they often are. Just think how many techniques and skills you already mastered. This next one you need to acquire will likely be no different. Do not pay too much attention to people that try to convince you that things will not work, especially if they don't specifically explain why that would be the case. **(Do this of course with carefully assessment of safety concerns).** Make a careful plan, get advice from open-minded experts and move ahead.

5. **Curate knowledge.** Create documents and, especially, videos of all the common laboratory actions. How to make an electrode, set up a electrolyzer cell (with the items you actually have in lab), make an XPS measurement, print a poster, register for classes, etc. Curate all COMSOL programs, excel spreadsheets, Mathematica files. Organize data repositories, instrument guides, instrument setups, tutorials, data processing scripts, etc. Recognize and appreciate those who spend time doing this and offer to include them in your research. The shared group website should be consulted regularly as it includes an incredible amount of useful resources, including reference books, data, how-to documents, standard operating procedures, example paper writing tools, and more.

6. **Make it enjoyable and meaningful.** Despite being hard, graduate school has the potential to be a very positive experience. You have huge amounts of freedom and relatively little responsibility relative to later stages in your life (e.g. procuring project funding, managing employees, etc.). You have a unique peer group that largely shares and empathizes with your challenges. You have the freedom to (and should) choose activities that best prepare you for **your own career aspirations.** You can truly drive change in your local environment to make it as enjoyable as possible. If you don't see a path forward to make it rewarding, consider doing something else. *You are not a failure or not worth less because you don't have a PhD.* In fact, in all stages in life you need to be willing to change course and try something new if a given course is not making you happy.

7. **Foster a supportive and inclusive community for those from all backgrounds.** Diversity brings different backgrounds, perspectives, and experiences to the table that can lead to more innovative and creative solutions. Embrace diversity in lab and across the institutions. Take intentional steps to create an environment that welcomes all groups of people. Refrain from using derogatory terms or statements that are harmful or disrespectful to others, avoid stereotyping or generalizations, educate yourself about common (often unintentional) discriminatory acts in the workforce, and be aware of personal implicit biases that may contribute to an exclusionary environment. Speak up for yourself and others if you see behavior inconsistent with the above.

8. **Celebrate success.** Each paper published is a real milestone, regardless of where it is published and what the impact factor of the journal is. Earning a MS degree is a big deal. Getting a job offer is a big deal. Passing each stage of your PhD process is important. Cultivate a group culture of celebration. Getting accepted to give talks at conferences, getting internships, getting honorable mention for awards, etc. – there are many small "wins" in graduate school. **Celebrate your and other's wins.**

9. **Fail fast… but learn from it.** Most ideas your advisor will provide *will not be well thought out.* Even the most well thought out ideas will "fail" as originally envisioned. But, you should make sure to learn something about why experiments didn't go as anticipated. **No experiment is a true failure unless you don't learn anything!** Be aware of what nature is telling you; and work with nature not against it. Doing experiments right and collecting high quality data is important (especially at the publication stage), but don't let perfection inhibit progress. Believe in and enjoy the process; the results **will** follow.

10. **Innovate to improve efficiency.** Group members should keep abreast of new technologies that could expedite processes for the group. A process that wastes a few minutes of everyone's day for years is worth spending time to redesign/improve. Group members should consider what routine processing steps can be expedited (for example with a script). Are you spending 30 minutes a day walking between two labs to access instruments X and Y? Maybe someone knows how to configure one of those instruments for remote access. Sometimes it is worth considering each individual step of a data workup and asking yourself, "is there a hot key for this operation…" If you get good at time saving shortcuts/hotkeys in a heavily used program, then present your findings as a tutorial or write a "how to" guide. With (widely) distributed lab space, how can you help other lab members to maximize research efforts? Be a good superuser, etc.

11. **Generate quality science hypotheses to test.** Generate research plans that focus on answering a question as opposed to making a "widget" or simply increasing performance (like a more active or stable AEMWE device). If you answer a reasonable question, *both the "yes" and "no" answer is interesting and publishable.* If you can't make a widget, the result is not publishable. Widgets are good outcomes, but usually it is better if they are **derived from a science advance and don't drive the science.** (i.e. if you figure out how a catalyst works, it is usually much easier to make it better!)

12. **Maintain high standards for your research.** Work to understand the equipment you use, how it works, and how the data is manipulated. Of course, do not fabricate or manipulate data without reporting how the data has been manipulated. Show all data to your advisor and discuss data to be discarded and the reasoning for this.

13. **Demand high safety standards from yourself and others.** Accidents happen too often in academic laboratories and can be very dangerous. Learn from near-misses, even small ones, and discuss these as a group and curate the learning about common mistakes and issues. There is no shame in making a mistake. Trust your instinct; if you see something that looks odd, ask somebody. Be kind but firm when correcting or advising on the practices of others. When receiving such feedback, don't take it personally. Realize that mistakes are bound to occur occasionally. They usually are pointing it out to make sure things are done safely and responsibly, not to belittle you or your ability. Keep the lab clean and clean up after yourself. However, don't be paralyzed. Learn how to do stuff safely, efficiently, and with confidence. Everyone has "work stop" authority when it comes to safety. If you see something unsafe, it is better to stop and reevaluate with advisor input (feel free to send text or call for immediate help), than risk it. Do not use AI tools for safety information, except to find original safety documents.

14. **Read literature in and outside your field.** Broaden your research horizon. You will not be doing what you were doing during your PhD for the rest of your life. Start with related fields. If you work on photoelectrochemistry, read some reviews from the solar cell community. If you work on electrocatalysis, read papers in the thermal heterogeneous catalysis field. Do not worry if you cannot understand all the specifics. Often, these are not well described in research articles in a language that is understandable for outsiders. Make sure you also cover relevant fields from an industrial perspective. With a PhD, you can be expected to understand how a field-effect-transistor works, even though you have never touched a semiconductor. A good place to start reading literature that is related (but not

identical) to areas of interest of our team is the Boettcher group RSS literature feed: https://zapier.com/engine/rss/5184145/Boettcherlab/ Some recommend inoreader for literature aggregation, a good interface on phone and computer.

15. **Conduct yourself professionally when representing the lab.** Speak and email with respect when communicating with industrial and academic representatives. Cultivating good first impressions can lead to interesting and productive collaborations and relationships down the road. You never know who or when your network will be critical to your professional goals. Do your research ahead of time to be prepared for meetings and discussions with collaborators. Be an example of this behavior for your mentees and new graduate students. Avoid negative or "trash" talking others in conversations with people you only have a professional relationship with. Consider the consequences of speaking too casually or "honestly" in a professional setting. It's hard to rebuild a bridge that you casually burnt.

16. **Use AI as a tool, but own your intellectual contributions:** Using AI for email templates, coding, and other tasks is encouraged as it improves work efficiency. AI can be an incredible search engine, but ask for sources **and be sure to know where your information is coming from.** When writing a paper, *you must own every word that goes into the publication*, and stay aware of journal stances on AI and authorship.

## C. Leadership & Vision

1. **Cultivate leaders.** Leaders are defined by their actions, not by their titles, and it is never too early in your career or studies to take leadership actions. Self-organize into smaller sub-units. Each sub-unit needs peer and near-peer leadership. Each sub-unit should meet on a weekly or biweekly basis to present progress and plan new experiments, (without the PI too). Each sub-unit could consider curating a journal feed with interesting new articles that are discussed and dissected and serve to help originate ideas. Senior graduate students and postdocs should see the value in mentoring younger students (see collaboration point above). Consider exploring options for leadership training outside of the research group.

2. **Practice ideation.** Try to generate new research ideas regularly. Discuss these with your sub-group. Think about basic science impact (i.e. new foundational knowledge) in addition to meeting possible applied technology goals. Both are important, but be deliberate. Practice "practical" and "big picture" ideation. Implement good ideas and/or discuss them with colleagues. Practice "what if" thought experiments in your head or with others. Attend talks/seminars. Ask yourself during seminars "how does this connect to my work" "how could I leverage this to gain insight in my system" "how does this connect to my colleagues' work"…. etc. **(always formulate a good question to ask at the end of the talk!).** Don't be afraid to develop your own research projects that you explore without your PI's knowledge (assuming it is not too expensive and can be done safely). Moonlight projects help you understand what it takes to independently ideate and execute. They are an exercise in evaluating your own creative limits and help you empathize with challenges faced by PIs. Your creative instincts are unique and will lead you to variable spaces that your PI may not have considered. Sometimes "failing" in the right direction is better than your advisor telling you not to do the experiment in the first place. *However, realize that you must balance trying your own ideas with executing and developing on advice provided by your advisor.*

3. **Give tutorial talks and idea pitches.** Tutorials (that are useful to group members) or new project pitches/proposals are excellent contributions to group meetings if well done.

4. **Create a culture of self-improvement.** You have access to incredible physical and intellectual resources. Use them. Make it normal for students to reach out to other groups to ask how to do things, for advice on experiments, etc. Reach out to alumni for advice. Leverage staff. Come to your advisor

with a list of specific questions **and be assertive in getting what you need to be successful.** Create a support structure in the group so that folks who are naturally better equipped to find collaborators and seek outside advice, advocate and support those who struggle more with this. Be a "connector" if you enjoy that. Remember, science is rarely done by individuals; it is done by teams. Leadership, mentorship and management of groups and individuals is critical to the success of scientific endeavors. Don't be afraid to ask questions from anyone in science, engineering, industry!

5. **Seek deep constructive criticism.** Practice giving and receiving constructive criticism. Deliver constructive criticism from a place of kindness, and work to get your own ego out of it. Receive constructive criticism with grace and sincerity, avoid a defensive stance. Constructive criticism and deep questioning often come off as aggressive and negative, but ultimately are extremely helpful in driving your project. *But,* also don't be afraid to communicate if you need a different "style" to receive that criticism. And be mindful of how your criticism is being received by others – it takes time to adapt to a mindset that accepts and appreciates such feedback and sometimes you should wait to deliver suggestions until the situation is appropriate. This openness and honesty applies everywhere, including safety, research, community, inclusivity, etc.

6. **Find a mentor who is not your research advisor.** Even if you only email with this person or have the occasional phone call to pick their brain, having an outside perspective can be invaluable. Look for another PI in the department, or elsewhere, who you respect and admire; reach out to former group members who have interesting careers; talk to the postdocs in your group and other groups. *More data is always better.* Reach out to group alumni, especially if seeking advice on work after graduate school. If you are struggling to find someone in your department who you feel would be a good mentor given your background and aspirations, reach out to someone at another university or institution or to your advisor/peers who could help find others in the community with whom you can develop a mentee relationship. Social media (Twitter, LinkedIn, etc.) can be useful to augment direct interactions.

## II. Joining the Boettcher Laboratory

The research group you join as part of your graduate studies is one of the most consequential decisions you will make in your educational career. The students chosen to join the group also have substantially consequential influence on the research, direction, intensity, and overall success and impact of our work. The financial costs of supporting PhD students at UC Berkeley and Lawrence Berkeley National Lab is substantially high, roughly $120,000 a year in 2025. Therefore, decisions to join a research group both from the student perspective and from the PI perspective are serious and done with utmost thought and consideration.

In addition to funding constraints that govern the number of students able to join the group each year, Professor Boettcher looks for a number of features in any PhD candidates for the group.

**Evidence for a "growth mindset" in research and career.** This is the most important aspect Prof. Boettcher assesses. This means that students enjoy and seek out opportunities to improve as scientists/engineers, communicators, educators, and innovators. This means students seek out honest *constructive criticism from the PI and others* (and be prepared to receive this honest criticism despite how difficult it may be to hear). **It means that students constantly seek to get better by acting on the feedback.** Those that enjoy engaging and thinking about their scientific and engineering studies in a rigorous and personally motivated way – including discussing your research within the group and with others and your advisor often.

**Evidence for commitment to research and a high level of work ethic.** No one is expected in the Boettcher laboratory to work any particular set of hours. The expectation is that everyone works hard and deeply cares about their work. This means the work in your PhD must be a very high priority and dedication to that is important. First-year students have a large number of commitments to classes, teaching, etc. Students are often engaged in many activities outside of the core of their PhD research. These can be valuable for your career, including seminars, workshops, business school innovation trainings, conferences, and more. But the

core passion and energy for the research and data analysis and product generation from that research is essential.

**Evidence for Intellectual Ownership.** A graduate degree at the PhD level is a personal journey that requires the candidate to be a full intellectual owner of a research project. Thus students are selected based on the promise that they will be able to, through mentoring and hard work, become scientists and engineers that are able to Identify knowledge gaps and create ideas, build project plans that test those ideas, and communicate those ideas; all while owning every step in the process with a high level of attention to detail. Of course, it takes time to get to this point – that is in fact the purpose of a PhD.

**Evidence for ability to support a positive group culture and ecosystem.** A positive, supportive, industrious, and innovative group culture and science ecosystem is central to group shared success. *Prof. Boettcher always get feedback from the research group on how interactions were prior to making a decision on whether to accept a particular PhD student.*

Professor Boettcher strives to take all the students that want join the group each year, but often this isn't possible financially or the fit is not ideal. In addition to the above, Professor Boettcher. will also consider interactions and performance in courses. While academics and innate "intelligence" (if there is such a thing!) are important, *much more important are the four points above*. Students in the group are sought that very much want to be there, get better at what they do, and are willing to put in the time and energy to succeed at a high level.

**The steps and process in learning about and joining the Boettcher Group.** In the Department of Chemical and Biomolecular Engineering, students submit their group preferences in mid-October. In the Department of Chemistry, students generally join groups between late September and mid-October. In the Material Science Department, students join groups prior to accepting their offer to join the department in April the year prior to arriving.

There are typically many students each year possibly interested in joining the Boettcher Laboratory. To create a fair and transparent process for students to learn about the group and convey their interest for joining, we have built the following set of events.

(1) Students interested in the laboratory typically email Shannon in the run-up to their arrival on campus in August or in the first few weeks after their arrival. Shannon will typically have Zoom or in-person meetings with students conveying significant interest in the research group.

(2) All students interested in exploring opportunities in the research group will be invited to a group event. This group event is typically two to three hours in length and often takes place at our laboratories at the top of Lawrence Berkeley National Lab in building 30 or 33. At this group event, various group members present research projects where there is possible opportunity for new students to contribute. We also discuss the group function and ecosystem and give some tours to our Berkeley lab facilities which complement the campus facilities. Shannon will happily sign anybody's green sheet for the chemistry department during this group evening or at another time.

(3) Following the group evening event, which complements department-sponsored poster sessions and open houses that may also occur, Shannon will typically meet one-on-one with students that remain highly interested in joining the group. Those students will also meet with group members working in areas they're interested to discuss research and understand the projects better. Interested students will often read recent publications from the group or may be able to peruse submitted or funded proposals governing the research projects.

**(4)** In September, Shannon will email those that remain interested in the group and ask them to schedule a formal interview. This interview will typically be in Shannon's office. The interview follows typical industry interview formats where there are questions about professional and career goals, technical background in research and coursework, and soft skills. Following the interview, Shannon may provide feedback on whether he thinks the fit is good in the group or not, either then or at a later time, depending on the number of students interviewing for positions.

(5) Students in the Chemical Engineering Department provide group preferences and those who are past the interview stage and invited to join the group are encouraged to indicate the Boettcher lab as their first or second group choice. For the chemistry department, arrangements can be directly made to join the group, following the conclusion of all the interviews.

# III. Various Online Accounts

### A. Group email list

The lab primarily communicates through email. The address Boettcherlab@lists.berkeley.edu will send an email to the full group. See the arrival checklist for subscription instructions. The group also uses slack, see website for details.

### B. Internal Website

The majority of lab information can be found on o UCB internal website (earlier see UO internal website). There are many resources here, such as the group meeting schedule and Zoom links (Group Calendar). If you are ever looking for information, first check the website. It is likely there somewhere.

### C. Microsoft Teams/Google Suite

The team uses google calendar and Zoom for meetings.

(UO: The lab has a Teams page with older information and files)

### D. Google Drive

All UCB documents should be organized in the google drive folder

### E. RSS Chemicals Inventory

UCB: Lab inventory is through RSS chemicals. Please make sure to tag chemicals as they come in with an RFID tag and remove them from inventory once done.

# IV. Meetings

### A. Group meeting

Our weekly group meeting itinerary is as follows:

1. Safety discussion + equipment discussion + other group business (<15 mins)
2. Literature, idea, or tutorial topic talk (~15 min of content- about 30 minutes total)
3. Research presentation (~30 min of content- About 1 hour talk)

Members (graduate students, visiting international scholars, postdocs, etc.) rotate giving topical and research presentations each meeting. Further details and the “batting order” can be found on the internal website under the Group Calendar tab.

At each group meeting, there is typically a group timekeeper that ensures discussion stays on course and excessive tangents are reined in to maintain flow and efficiency of the meeting.

When group meetings are in person snacks are provided. Contact the snack manager for requests (Duha, duhasyar@berkeley.edu ).

Group members also use group meetings to give practice talks. It is general practice to give a practice talk at least 1 week before any committee meeting or before any conference talk. Contact the scheduling manager to schedule any practice talk.

### B. Subgroup

The group projects are organized into teams. Each subgroup meets either weekly or biweekly. Meeting times and links can be found on the internal website under the Group Calendar tab.

*Any data update and presentations should be immediately uploaded to a folder on the Google Drive Folder*
Shannon frequently uses the data from these updates when working on grant proposals, conference talks, etc., so it is important this is done consistently and in a timely manner.

### C. Individual meetings with Shannon

Some members prefer to meet individually with Shannon to discuss data and upcoming experiments. If interested, discuss directly with Shannon. His Calendly link is:

## V. Absences

### A. Illness

If you are sick, you should generally try to work (mostly) from home when not resting up to get better. If you have scheduled meetings you can not attend, let those attending, particularly your PI, know via an email. If you are responsibilities that are central to group function, particularly related to grant reporting and presentations to program managers, make sure you get ahold of Shannon or other key researchers on the team such that these are taken care of.

### B. Vacation

Let Shannon know in advance if you are going to be gone for any extended period of time. There is generally no specific limit on absences/days off, however, it is expected all group members will work as hard and efficiently as possible to achieve their own goals when present, and take reasonable time off to recharge or otherwise deal with other commitments. Often significant work can be accomplished, for example writing, reading, during travel portions of trips etc. Students and postdocs are encouraged to balance their efforts as they see most productive and appropriate for their individual situation and consistent with their professional and personal goals.

Shannon will generally discuss with you in person if he feels research progress is for any reason not progressing in a way consistent with achieving a PhD or other professional goals.